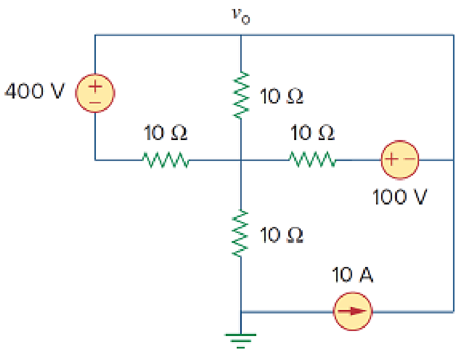
Chapter 3, Problem 51P

Problem

Apply mesh analysis to find *vo* in the circuit of Fig. 3.96.

Figure 3.96:



Step-by-step solution

Step 1 of 4

Refer to Figure 3.96 in the text book.

Identify the loops and currents in the circuit diagram shown in Figure 1.

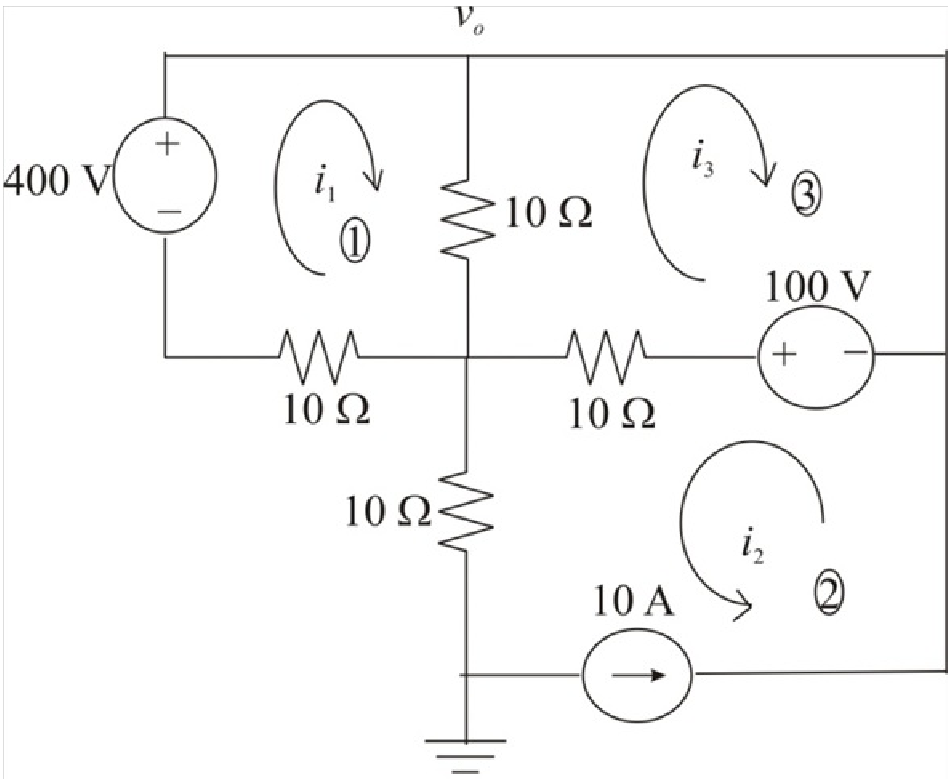


Figure 1

Step 2 of 4

Notice that the current i_2 is, $10\ \text{A}$.

Apply mesh analysis in loop 3.

Apply Kirchhoff's voltage law, the algebraic sum of all voltages around a closed path (or loop) is zero.

$$-100+(i_2+i_3)(10)+(i_3-i_1)(10)=0$$
$$-100+10i_2+10i_3+10i_3-10i_1=0$$
$$-100+10i_2+20i_3-10i_1=0$$
$$-10+i_2+2i_3-i_1=0$$

Substitute $10\ \text{A}$ for i_2.

$$-10+10+2i_3-i_1=0$$
$$2i_3-i_1=0$$
$$i_1=2i_3$$

Therefore, the expression for the current i_1 is, $2i_3$.

Step 3 of 4

Apply mesh analysis in loop 1.

For mesh 1, Kirchhoff's voltage law gives,

$$i_1(10)-400+(i_1-i_3)(10)=0$$
$$i_1-40+i_1-i_3=0$$
$$2i_1-40-i_3=0$$

Substitute $2i_3$ for i_1.

$$2(2i_3)-40-i_3=0$$
$$4i_3-40-i_3=0$$
$$3i_3-40=0$$
$$3i_3=40$$
$$i_3=\frac{40}{3}$$
$$=13.33\ \text{A}$$

Therefore, the value of the current i_3 is, $13.33\ \text{A}$.

Step 4 of 4

Calculate the value of voltage v_o.

$$v_o = i_2(10) + (i_1 - i_3)(10)$$

Substitute $2i_3$ for i_1.

$$\begin{aligned} v_o &= i_2(10) + (2i_3 - i_3)(10) \\ &= i_2(10) + i_3(10) \\ &= (i_2 + i_3)(10) \end{aligned}$$

Substitute 13.33 A for i_3 and 10 A for i_2.

$$\begin{aligned} v_o &= (10 + 13.33)(10) \\ &= (23.33)(10) \\ &= 233.3 \text{ V} \end{aligned}$$

Therefore, the value of the voltage v_o is, $\boxed{233.3 \text{ V}}$.

Comments (2)

Anonymous

How on earth does the voltage in the top and bottom 10 ohm resistors equal to V_0?

Anonymous

Because if you measure the voltage across point V_0 and ground then you are measuring the drop across those middle resistors only

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